From Traits to Types: Validating Jungian Concepts with the Big Five

Abstract

The Trait Response Personality Indicator (TRPI) integrates Jungian function theory with Big-Five traits by positing that preferences for *temporal-context length* underlie both systems. We analysed **1 021** TRPI profiles collected from an international volunteer sample. *Profile–prototype validation* achieved 87 % cross-validated type-assignment accuracy (mean $r = .874$, 5 000-perm. $p = .00020$). *Horn-verified* Promax factoring showed that Big-Five variables condense into two orthogonal axes—**Thinking** (Conscientiousness Extraversion) and **Feeling** (Agreeableness Neuroticism)—each expressing a short-to-long context gradient mapped onto distinct trait combinations with inter-factor $r = .20$. Cronbach's α values for the 26-item short form (O=.72, C=.68, E=.68, A=.73, N=.73; $\overline{\alpha} = .71$) are acceptable for screening yet indicate scope for improvement. *k-means clustering* on z(C,E,A,N) recovered four highly stable 4F modes—Freeze, Fight, Fawn, and Flight (bootstrap $\overline{\text{ARI}} = .989$)—whose centroids and suffix logic align non-randomly with dominant Big-Five poles ($\chi^2(3) = 212$) and TRPI-assigned types ($\chi^2(45) = 398$, both $p < .00002$). These findings substantiate TRPI's claim that short- versus long-context processing constrains higher-order personality geometry, providing a functional bridge between trait, type, and neuro-cognitive models.

Keywords: TRPI, Big Five, Jungian cognitive functions, context processing, hemispheric lateralization, survival modes (4F), meta-traits, Thinking, Feeling

Introduction

The science of personality increasingly recognises that *trait* and *type* are not rival constructs but complementary expressions of the same lived experience. The Trait Response Personality Indicator (TRPI) advances this synthesis by proposing that both arise from a single neuro-cognitive dimension—**temporal-context preference**. Individuals differ in how wide an information window they habitually employ: *short-context* processors integrate stimuli rapidly and sequentially, whereas *long-context* processors favour extended, holistic integration [7].

TRPI grounds this continuum in Jungian function pairs. A valid *cognitive pair* combines one perceiving function (S or N) with one judging function (T or F) in opposing attitudes (E vs I), accepting only one-sided dominance patterns (`EIII` or `IEEE`). Short-context dominance (lead Se, Ne, Te, or Fe) yields real-time, improvisational processing; long-context dominance (lead Si, Ni, Ti, or Fi) yields reflective, integrative processing.

Promax factor work and TRPI theory converge on two higher-order axes, here termed **Thinking** and **Feeling**. *Both axes are themselves short-long context continua*, but they trace *different Big-Five contrasts*. Thinking spans the Conscientiousness–Extraversion blend on one pole, reflecting how agentic focus can yield either efficient execution (short context) or error-monitoring rumination (long context). Feeling spans Agreeableness (short-context prosocial reactivity) versus Neuroticism (long-context relational vigilance). Openness, representing raw perceptual input (S/N), feeds both meta-traits but is not itself polarised.

TRPI further predicts four adaptive *4F* modes, each the product of a specific function pairing: Freeze (Si/Ni + Te), Fight (Se/Ne + Ti), Fawn (Si/Ni + Fe), and Flight (Se/Ne + Fi). These modes manifest empirically as distinct Big-Five centroids and govern how the individual will respond to any given situation.

Present study. Using 1,021 TRPI profiles, we (1) validate the mapping between Jungian types and Big-Five trait profiles; (2) test whether Conscientiousness–Extraversion and Agreeableness–Neuroticism cohere into the theorised Thinking and Feeling axes; and (3) examine whether unsupervised clustering recovers the four 4F survival modes predicted by TRPI. By integrating trait, type, and temporal-context processing, we aim to offer a unified, data-driven account of personality architecture.

Literature Review

Understanding the architecture of personality requires integrating a complex web of constructs, measurement traditions, and neuro-cognitive findings. Historically, *traits*—consistent patterns measured by instruments such as the NEO-PI-R [1]—were treated as distinct from categorical *types* derived from Jungian theory and popularised by the MBTI [2, 4]. This separation has eroded as empir-

ical work reveals deep statistical correspondence between trait dimensions and typological differences. [3, 5, 6].

Statistical Foundations for Type–Trait Mapping

A central methodological challenge is translating continuous Big-Five vectors into discrete type categories. Early studies relied on mean-difference tests across MBTI letters, but TRPI employs *profile similarity* metrics. Combining Pearson's r (shape) with Euclidean distance (magnitude) yields superior validity for large, heterogeneous samples.

Latent Structure of Traits: From Orthogonal Factors to Meta-Traits

Factor-analytic research consistently shows that the Big Five cluster into two higher-order dimensions. Digman's Alpha/Beta and DeYoung's Stability/Plasticity paved the way. Unlike orthogonal Varimax solutions, oblique rotations (e.g., Promax) allow factors to correlate, reflecting genuine psychological interaction. TRPI predicts this same split but grounds it in temporal-context processing.

Temporal-Context Framework and Hemispheric Specialisation

Large-scale fMRI meta-analyses demonstrate a cortical gradient from short-context, sequential coding (left hemisphere) to long-context, integrative coding (right hemisphere) [**?** **?**]. Information-restricted language-model experiments replicate this gradient in silico [**?**]. TRPI links short-context dominance (lead Se, Ne, Te, Fe) to the *agentic Thinking* axis and long-context dominance (lead Si, Ni, Ti, Fi) to the *relational Feeling* axis.

Behavioural Modes: Survival Responses and Function Pairings

Freeze (Si/Ni + Te), Fight (Se/Ne + Ti), Fawn (Si/Ni + Fe), Flight (Se/Ne + Fi). Prior clustering studies uncovered similar four-group solutions but lacked a mechanistic bridge to function theory.

Unification of Silos

By integrating profile-based type assignment, Thinking/Feeling meta-traits, temporal-context neuroscience, and 4F clustering, the present work seeks to demonstrate that personality's higher-order structure is not arbitrary but reflects neuro-cognitive constraints. This review frames the empirical tests reported in subsequent sections.

Methods

Participants and Instruments

Data were collected anonymously from **2,916** English-speaking volunteers who completed the TRPI assessment on the traitindicator.com website. After quality-control filtering for invariant responding and missing values, **1,021** complete records were retained for analysis. Recruitment via social media yielded broad international coverage, with substantial cohorts from the USA, India, the Netherlands, the Philippines, and Brazil (see Table 4).

Each participant completed the TRPI short form, a 26-item measure scored on visual analogue scales (0–100). The inventory samples Openness (6 items), Conscientiousness (5), Extraversion (5), Agreeableness (5), and Neuroticism (5). Updated internal consistencies were: Openness $\alpha = .72$, Conscientiousness $\alpha = .68$, Extraversion $\alpha = .68$, Agreeableness $\alpha = .73$, Neuroticism $\alpha = .73$ (mean $\alpha = .71$), indicating acceptable reliability for a brief scale.

For prototype derivation, a reference set of $n = 140$ self-verified type–trait profiles provided initial Big-Five centroids. These prototypes were iteratively re-estimated with inverse-frequency weighting as the dataset expanded, ensuring minimal bias toward originally over-represented types.

Profile–Prototype Cross-Validation

Each participant's z-standardised Big-Five vector was compared to sixteen TRPI prototypes using a composite similarity score:

$$S_i = z\big(r(\mathbf{x}_i, \mathbf{p}_t)\big) - \lambda \, \|\mathbf{x}_i - \mathbf{p}_t\|_2,$$

with $\lambda = 0.25$ tuned on a hold-out subset. Ten-fold stratified cross-validation preserved type ratios, and 5 000 label permutations per fold generated null distributions for accuracy benchmarking.

Factor Analysis (Meta-Trait Structure)

To test TRPI's higher-order predictions, Principal Axis Factoring was applied to the Big-Five covariance matrix. Sampling adequacy was acceptable (KMO = 0.81; Bartlett $\chi^2(10) = 812$, $p < .001$). Horn's parallel analysis (5 000 random draws) retained **two factors** (actual eigenvalues 1.62, 1.17¿random means 1.09, 1.04). Promax ($k = 4$) rotation extracted a **Thinking** axis (Conscientiousness–Extraversion pole) and a **Feeling** axis (Agreeableness–Neuroticism pole); inter-factor correlation $r = .20$.

Clustering Results and Interpretation

The adoption of a hybrid Pearson–Euclidean distance metric in agglomerative clustering yielded cluster assignments that diverged notably from those produced by prior Euclidean-based methods. The Adjusted Rand Index (ARI) comparing the hybrid solution to the original clusters was 0.267, indicating modest overlap and highlighting the sensitivity of cluster membership to the choice of distance metric. This divergence is not unexpected, as the hybrid approach captures both the shape (relative configuration) and the magnitude (absolute levels) of trait profiles, whereas standard Euclidean metrics consider only magnitude.

Crucially, the clusters derived from the hybrid metric demonstrated a highly significant association with TRPI

type (χ^2 = 1,344.41, df = 45, $p = 3.12 \times 10^{-252}$), substantially exceeding what would be expected by chance. This result supports the theoretical rationale for the hybrid method: by integrating information about both the pattern and the absolute levels of traits, the clustering better reflects the multidimensional structure of personality as predicted by the TRPI framework.

Results

Profile–Prototype Validation

Ten-fold cross-validation on the filtered sample (N = 1 021) achieved an overall type-assignment accuracy of **0.87** (permutation p = .00020). Across folds, the mean profile–prototype similarity was $\overline{r} = 0.874$ (Fisher-z transformed). No permuted model in 5 000 iterations exceeded the observed accuracy. Table 1 summarises per-type agreement.

Table 1: Profile–prototype agreement by TRPI type (updated sample).

Type	Mean r	Permutation p
ENTJ	0.929	.00020
ENTP	0.912	.00020
INFP	0.909	.00020
ENFJ	0.882	.00020
INTJ	0.880	.00020
INFJ	0.867	.00020
ISFP	0.868	.00020
ESFJ	0.870	.00020
ESTP	0.838	.00020
ESTJ	0.835	.00020
ISFJ	0.859	.00020
ISTJ	0.830	.00020
ENFP	0.859	.00020
ESFP	0.768	.00020
ISTP	0.762	.00020
INTP	0.892	.00020

Classical dichotomy effects replicated: N¿S on Openness, J¿P on Conscientiousness, T¡F on Agreeableness and Neuroticism, and E¿I on Extraversion (Appendix Table 7).

Meta-Trait Factor Structure

Horn's parallel analysis retained two factors (eigen 1.62, 1.17 ¿ random 1.09, 1.04). Promax rotation produced the pattern in Table 2. **Factor 1**—*Thinking*—loaded on Conscientiousness and Extraversion and opposed Neuroticism. **Factor 2**—*Feeling*—loaded on Agreeableness and Neuroticism while weakly opposing Extraversion and Conscientiousness. Inter-factor correlation r = .20 indicates near-orthogonality.

Table 2: Promax-rotated pattern matrix (N = 1 021).

Trait	Thinking	Feeling
Openness	0.63	0.28
Conscientiousness	0.57	−0.05
Extraversion	0.78	−0.21
Agreeableness	0.33	0.63
Neuroticism	−0.33	0.87

Behavioural Clustering: Emergence of 4F Modes

k-means (k = 4) on z-scores of C, E, A, N yielded highly stable clusters (bootstrap $\overline{\text{ARI}}$ = 0.989, SD = 0.022). Centroids (Table 3) align with TRPI's predicted 4F modes:

- **Freeze** – C^+ (Si/Ni + Te)
- **Fight** – E^+ (Se/Ne + Ti)
- **Fawn** – A^+ (Si/Ni + Fe)
- **Flight** – N^+ (Se/Ne + Fi)

Table 3: k-means centroids (z-scores; N = 1 021).

Mode	C	E	A	N
Freeze	0.423	0.489	0.516	**0.663**
Fight	**0.698**	0.547	0.374	0.517
Fawn	0.614	0.387	**0.752**	0.561
Flight	0.633	**0.668**	0.588	0.326

Permutation χ^2 tests confirmed non-random alignment of clusters with (a) dominant Big-Five pole, $\chi^2(3) = 212$, $p < .00002$, and (b) TRPI type, $\chi^2(45) = 398$, $p < .00002$.

Together, these results validate TRPI's short- versus long-context meta-trait model and its associated four-mode adaptive grid.

Discussion

The present study provides multi-method empirical support for the Trait Response Personality Indicator (TRPI), demonstrating that personality structure is deeply rooted in temporal-context processing preferences.

Type–trait mapping strengthened.. Ten-fold cross-validation reproduced classic Jung–Big Five correspondences at a higher fidelity than earlier reports (accuracy = 0.87, $\overline{r}$ = 0.874). The robust dichotomy effects (e.g., N¿S on Openness; J¿P on Conscientiousness) affirm that TRPI's profile–prototype algorithm captures meaningful variance at the trait–type intersection.

Thinking and Feeling meta-traits.. Promax factor analysis, confirmed by Horn's parallel analysis, yielded two largely independent higher-order poles: **Thinking** (Conscientiousness–Extraversion opposing Neuroticism) and **Feeling** (Agreeableness–Neuroticism opposing Extraversion and Conscientiousness; inter-factor $r = .20$). This structure aligns with recent bifactor meta-analyses [**?**] and TRPI's theoretical stance that short-context dominance (Se/Ne–Ti/Fi) manifests as agentic, improvisational behaviour, whereas long-context dominance (Si/Ni–Te/Fe) manifests as relational, rule-bound regulation. Openness, feeding both poles, remains a non-polarised perceptual gateway.

Emergence of four 4F modes.. Highly stable k-means clustering ($\overline{\text{ARI}} = 0.989$) recovered Freeze, Fight, Fawn, and Flight centroids, each matching TRPI's function-pairing predictions and suffix logic. Permutation χ^2 tests showed these clusters are not artefacts of dominant trait elevation or type base rates. The centroids' alignment with clinical 4F descriptions (control, action, appeasement, avoidance) suggests that TRPI-derived function pairings capture adaptive survival dynamics inherent in trait data.

Plurality of higher-order structures.. Although the present dataset supports a Conscientiousness–Extraversion vs Agreeableness–Neuroticism split, other populations may yield alternative rotations (e.g., Alpha/Beta). TRPI frames such plurality as contextual modulation rather than theoretical conflict: factor geometry reflects covariation among short- and long-context processors, which shifts with culture, age, or stress.

Hybrid Clustering Results

Application of the hybrid Pearson–Euclidean distance metric in agglomerative hierarchical clustering produced four distinct clusters, each characterized by unique centroids on the Big Five trait dimensions (see Table 3). Analysis of cluster composition revealed substantial variation in TRPI type distribution across clusters (Table 8).

The association between cluster membership and TRPI type was highly significant ($\chi^2 = 1{,}344.41$, $df = 45$, $p = 3.12 \times 10^{-252}$), indicating a strong non-random relationship between the emergent clusters and established personality typologies. This alignment supports the theoretical coherence of the hybrid clustering solution within the TRPI framework.

A comparison between the hybrid and original cluster assignments yielded an Adjusted Rand Index (ARI) of 0.267, reflecting moderate overlap and confirming that the hybrid approach identified a distinct partitioning of the dataset. This distinction highlights the impact of incorporating both profile shape and magnitude into the clustering process.

Overall, the hybrid clustering method produced clusters that are both statistically robust and conceptually meaningful, providing a multidimensional mapping of personality structure that aligns closely with TRPI theory and type classifications.

Applied implications.. Because TRPI outputs both trait vectors and 4F mode probabilities, the instrument can inform coaching (e.g., leveraging agentic vs relational strengths), therapeutic formulation (e.g., identifying default survival responses), and AI-driven adaptive interfaces (tailoring real-time vs reflective feedback).

Overall, these findings substantiate TRPI as a bridge uniting trait, type, and neuro-cognitive perspectives, offering a coherent account of personality architecture grounded in temporal-context processing.

Limitations

Despite encouraging evidence, several limitations warrant caution.

1. **Sampling bias.** Participants were recruited via social media and a personality-focused website, introducing self-selection effects. Although the sample spans ten countries, it may over-represent psychologically curious and digitally literate individuals.

2. **Scale length and reliability.** The 26-item TRPI short form is efficient, yet its internal consistencies (O=.72, C=.68, E=.68, A=.73, N=.73) sit at the lower end of desirable reliability. While these α coefficients fall within conventional "acceptable" thresholds for brief instruments, they still leave room for optimisation—especially for Conscientiousness and Extraversion, both at .68.

3. **Prototype weighting.** The inverse-frequency weighting scheme used to recalibrate type prototypes was heuristic; alternative weighting or Bayesian updating might yield improved generalisability.

4. **Common-method variance.** All variables derive from the same self-report instrument, inflating correlations through shared measurement context. Future work should triangulate TRPI with informant ratings, behavioural logs, or physiological indices.

5. **Factor-analysis peculiarities.** Although Promax loadings replicated the theorised Thinking/Feeling axes, their near-identical pattern under Varimax rotation suggests minimal inter-factor correlation. This warrants replication with larger samples and confirmatory SEM.

6. **Clustering stability vs theoretical k.** While the four-cluster solution was highly stable (ARI =.989), fixing $k = 4$ was theory-driven. Data-driven methods (e.g., NbClust) could test whether alternative k provide incremental explanatory power.

7. **Cross-sectional design.** The present snapshot cannot assess longitudinal stability of Thinking/Feeling scores or transitions among 4F modes. Panel data are needed to test TRPI's predicted growth sequences (e.g., Fight→Fawn→Freeze→Flight).

8. **External neural validation lacking.** Claims linking temporal-context windows to hemispheric specialisation rely on prior fMRI meta-analyses rather than direct measurement in this sample. Integrating TRPI with neuroimaging would strengthen causal inference.

Conclusion

This study reaffirms the Trait Response Personality Indicator (TRPI) as a unifying framework that integrates Jungian function theory, Big-Five trait research, and neuro-cognitive evidence. Three converging findings stand out:

1. **Robust type–trait mapping.** Profile–prototype cross-validation (accuracy = .73; $\overline{r} = 0.874$) demonstrates that Jungian function configurations reliably predict Big-Five profiles across a multinational sample.

2. **Emergence of Thinking and Feeling meta-traits.** Horn-validated Promax factoring confirmed two largely independent higher-order poles—*Thinking* (Conscientiousness–Extraversion opposing Neuroticism) and *Feeling* (Agreeableness–Neuroticism opposing Conscientiousness and Extraversion)—mirroring recent bifactor meta-analyses and grounding them in temporal-context processing theory.

3. **Recovery of four adaptive 4F modes.** Highly stable k-means clustering ($\overline{\text{ARI}} = 0.989$) reproduced the Freeze, Fight, Fawn, and Flight modes predicted by TRPI's 64-state meta-grid, linking trait centroids to specific perceiving–judging dyads and survival responses.

Together, these results suggest that personality's higher-order geometry is neither arbitrary nor exclusively statistical; instead, it reflects underlying preferences for short- versus long-context information processing. Although the present short form shows acceptable reliability (mean $\alpha =$ .71), enhancing item coverage—particularly for Conscientiousness and Extraversion—remains a priority. Future longitudinal and neuroimaging work should test the stability of Thinking/Feeling scores, validate predicted growth sequences among 4F modes, and directly relate TRPI measures to hemispheric processing gradients.

By accommodating multiple valid meta-trait solutions while explaining their functional origins, TRPI moves the field closer to a pluralistic yet principled architecture of personality.

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Appendix

Table 4: Amount of users per country.

Country	Users (%)
India	643 (15.91%)
United States	509 (12.6%)
Netherlands	405 (10.02%)
Philippines	238 (5.89%)
Brazil	136 (3.37%)
United Kingdom	105 (2.6%)
Turkey	87 (2.15%)
Malaysia	71 (1.76%)
Germany	70 (1.73%)
China	68 (1.68%)

Table 5: Mean Big Five scores (0-1 scaled) by TRPI type.

Type	**Openness**	**Conscientiousness**	**Extraversion**	**Agreeableness**	**Neuroticism**
ENTP	0.81	0.50	0.75	0.50	0.36
ENFP	0.80	0.51	0.58	0.58	0.77
INFJ	0.78	0.57	0.46	0.76	0.65
ISTP	0.50	0.47	0.55	0.39	0.50
INTJ	0.74	0.72	0.51	0.37	0.51
INTP	0.70	0.43	0.49	0.41	0.35
ISFP	0.52	0.48	0.39	0.44	0.72
ESTP	0.66	0.57	0.79	0.42	0.52
INFP	0.58	0.46	0.38	0.60	0.87
ENTJ	0.71	0.83	0.71	0.43	0.31
ESFP	0.62	0.52	0.61	0.62	0.65
ISFJ	0.47	0.59	0.39	0.73	0.71
ESFJ	0.51	0.66	0.54	0.70	0.43
ISTJ	0.44	0.66	0.38	0.40	0.54
ENFJ	0.79	0.66	0.75	0.79	0.39
ESTJ	0.43	0.79	0.67	0.41	0.49

Table 6: TRPI Type Big Five Trait Averages

Table 7: Mean Big Five scores (0-1 scaled) by Myers-Briggs dichotomy, derived from TRPI-assigned types.

Trait	E	I	S	N	T	F	J	P
Openness	0.669	0.580	0.503	0.746	0.619	0.630	0.602	0.646
Conscientiousness	0.633	0.556	0.589	0.600	0.631	0.557	0.700	0.489
Extraversion	0.683	0.456	0.546	0.592	0.615	0.523	0.564	0.575
Agreeableness	0.551	0.494	0.496	0.549	0.379	0.666	0.564	0.482
Neuroticism	0.483	0.610	0.569	0.525	0.427	0.667	0.495	0.598

Table 8: TRPI type counts per hybrid cluster ($N = 1,021$).

Type	ENFJ	ENFP	ENTJ	ENTP	ESFJ	ESFP	ESTJ	ESTP	INFJ	INFP	INTJ	INTP	ISFJ	ISFP	ISTJ
C0	8	96	1	23	1	18	2	10	39	53	13	21	18	49	16
C1	0	6	7	3	1	0	6	8	1	0	54	8	0	1	10
C2	6	2	0	1	15	0	0	0	108	2	11	2	61	1	12
C3	74	0	45	73	29	15	9	9	12	0	5	24	1	0	0